Appendix 4: Meta-analysis and other calculations

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1 FURTHER DETAILS OF METHODS

The protocol was first registered on PROSPERO CRD42020182044.[1] Following feedback from our content experts who at that stage were blinded to the search results, amendments were made pre-data extraction and a revised protocol was submitted for publication (Supplementary file: protocols).[2] This included expanding the inclusion criteria from only including RTIs that can be caused by a coronavirus to RTIs caused by any virus (e.g. rhinovirus inoculation), tightening the exclusion criteria to exclude respiratory illnesses not predominantly caused by viral infections (e.g. pneumonia in adults) unless a viral infection is confirmed, and finalising the measures of effect to be extracted for the *a priori* outcomes. For pragmatic reasons, the *post hoc* decision was made to only extract data for adult populations.

RevMan 5.4,[3] R software,[4 5] Microsoft Excel, and GRADEpro GDT[6] were used for the statistical analyses. Irrespective of statistical heterogeneity, due to considerable clinical and methodological heterogeneity random effects models were used for all meta-analyses.

The effect measures for dichotomous outcomes were risk ratios, calculated using the Mantel-Haenszel method. Events measured over different timeframes were calculated and reported as the incidence rate per person-months (Table 6.6), from which incidence rate ratios (IRR) and rate differences (RD) were estimated using an inverse variance method. Studies reporting separate counts for different types of viral RTIs (e.g. common cold, bronchitis, flu-like illness) were combined to calculate the total number of RTIs. In one RCT there were no moderate RTI events in the zinc arm.[7] In another RCT there were no adverse events (AEs) in the zinc arm.[8] In both instances, it was not a rare event, the study arm sizes were balanced, there was only one zero cell in the meta-analysis, and meta-analysis heterogeneity was acceptable. Therefore, as per Cochrane guidance, 0.5 was recorded to facilitate analysis.[9]

For continuous data, either the weighted or standardised mean differences were calculated using the generic inverse variance method. Day-3 mean symptom severity scores were transformed to a modified Jackson common cold scale (Table 6.2).[10] For the secondary outcome, mean days duration, data was not imputed and instead extracted from published systematic reviews who had performed the calculations or contacted the authors (Table 6.5).[11-13] For the 4-arm RCT (Table 4.2), as per the preferred approach recommended in the Cochrane Handbook,[9] the three zinc arms were combined and the SD was calculated with RevMan. To allow comparisons of zinc doses and salts, the number of participants in the placebo arms were divided into thirds. A sensitivity analyses compared the two approaches (Table 4.2, Analysis 4.2.7). Rapid review constraints included not imputing missing data for any secondary outcome such as mean days duration, and not

contacting the authors. Instead, additional information from previous systematic reviews was extracted (Table 6.5).[11-13]

For time-to-event data, hazard ratios (HR) were calculated using an O-E variance method. Data extracted from survival curves with WebPlotDigitizer Version 4.2[14] (Table 6.5) was imputed for the first seven days using the direct method 10 in the 'HR calculations spreadsheet' published by Tierney et al.[15] The other option was the direct method 11. that used less of the data that was extracted. Sensitivity analysis confirmed the effect estimate was relatively stable and that method 10 was the more conservative approach (Table 4.1, Analysis 4.1.8). For the two 4-arm RCTs (Table 4.2), the three zinc arms were combined. It was not possible to accurately divide the daily survival counts for the placebo groups into thirds. This unaddressed correlation was ignored to allow comparisons of zinc doses and salts and a sensitivity analysis compared the two approaches (Table 4.2, Analyses 4.2.4 and 4.2.5.). The sensitivity analysis found an insignificant conflation from the unaddressed correlation arising from the unit-of-analysis error (Table 4.2, Analysis 4.2.7).

When data from at least 10 studies were pooled, publication bias was assessed with visual inspection of contour-enhanced funnel plots (Figures 1, 2, and 3) and statistically analysed using Egger's regression for continuous outcomes and the Harbord score for dichotomous outcomes.[16] Due to ongoing methodological uncertainties no statistical test was used for hazard ratio outcomes.[16 17]

A priori subgroup analyses were conducted for different ages groups, RTI causes and severity, and zinc administration routes, salts, and doses. Zinc doses were converted to milligrams (mg) of elemental zinc per day (Table 6.1). *A priori* subgroup analyses compared ages groups, RTI causes and severity, and zinc administration routes, salts, and doses. *Post hoc* subgroup analyses compared days symptomatic prior to study enrolment and study definitions of symptomatic recovery. To investigate potential dose effects of oral or sublingual zinc, the chi² test comparing three categories (<50mg daily, 50-200mg daily, >200mg daily) was used for dichotomous and time-to-event data. The categories were selected *post hoc* based on a no observed adverse effect level (NOAEL) of 50mg and a higher risk of more severe adverse effects, such as vomiting, with doses above 225mg.[18]

A priori sensitivity analyses investigated the point estimate and 95% CI changes when study outcomes with a higher RoB and statistical outliers were removed. *Post hoc* sensitivity analyses investigated the analytical decision zero with 0.5 for person-time rates (Analyses 2.1.7., 2.1.8., 5.2.6., 5.2.7.), the impact of reclassifying the control lozenge containing an unspecified amount of quinine as a placebo control (Analysis 4.1.9.) and combining the three zinc arms and dividing the placebo arms of the two 4-arm RCTs (Analysis 4.2.7).

Statistical heterogeneity was assessed using the Cochrane Q test and I^2 statistic that were interpreted according to guidelines from the Cochrane Handbook (0–40%: might not be important; 30–60%: may represent moderate heterogeneity; 50–90%: may represent substantial heterogeneity; 75–100%: considerable heterogeneity).[9] When statistical heterogeneity was identified, we explored possible explanations with subgroup and sensitivity analyses.

For symptom severity on day-3 for mild RTIs, the MID for mean difference was *post hoc* set at 1 point on a standardized scale (Analysis 6.3).[19 20] For the purpose of estimating an absolute effect from zinc use, the probability of remaining symptomatic on day-7 was set *post hoc* at 33% for the placebo and active controls.[21]

2 META-ANALYSIS RESULTS: PREVENTION

2.1 Pre-exposure prevention (1 – 12 months) of community acquired respiratory tract infections (RTIs): Zinc vs. placebo

Outcome	Studies	Participants (person months)	Statistical Method	Effect estimate [95% CI]	P value	I^2
2.1.1 Risk of any respiratory tract infection (RTI) per person-months						
Incidence RTIs [7 8 22 23]	4	2804 (3565)	IRR (IV, R)	**0.68 [0.58, 0.80]**	**<0.001**	0%
			IRD (IV, R)	**-0.05 [-0.08, -0.01]**	**0.01**	**63%***
2.1.2 Risk of a mild symptoms only (e.g. common cold) per person-months						
Incidence mild RTIs [7 22 23]	3	2767 (3306)	IRR (IV, R)	**0.72 [0.61, 0.85]**	**<0.001**	0%
			IRD (IV, R)	**-0.05 [-0.07, -0.02]**	**0.02**	7%
2.1.3 Risk of a moderately severe symptoms (e.g. flu-like illness) per person-months						
Incidence moderate RTIs [7 22 23]	3	2767 (3306)	IRR (IV, R)	**0.13 [0.04, 0.38]**	**<0.001**	0%
			IRD (IV, R)	**-0.01 [-0.02, -0.01]**	**<0.001**	0%
2.1.4 Risk of RTIs in low & high-risk age groups per person-months						
Incidence RTIs young adults [8 22 23]	3	2755 (2977)	IRR (IV, R)	**0.70 [0.59, 0.82]**	**<0.001**	0%
Incidence RTIs older adults [7]	1	49 (588)	IRR (IV, R)	**0.38 [0.16, 0.91]**	**0.03**	n/a
Test for subgroup differences: $Chi^2 = 1.91$, df = 1 (P = 0.17), $I^2 = 47.6\%$						
Incidence RTIs young adults [8 22 23]	3	2755 (2977)	IRD (IV, R)	-0.05 [-0.11, 0.00]	0.07	**79%***
Incidence RTIs older adults [7]	1	49 (588)	IRD (IV, R)	**-0.04 [-0.07, -0.01]**	**0.02**	n/a
Test for subgroup differences: $Chi^2 = 0.20$, df = 1 (P = 0.65), $I^2 = 0\%$						
2.1.5 Risk of RTIs according to zinc administration route and daily dose per person-months						
Incidence RTIs 45 mg oral [7]	1	49 (588)	IRR (IV, R)	**0.38 [0.16, 0.91]**	**0.03**	n/a
Incidence RTIs 15 mg oral [8]	1	30 (259)	IRR (IV, R)	1.06 [0.29, 3.96]	0.09	n/a
Incidence RTIs 1.2 mg topical nasal spray [22 23]	2	2718 (2718)	IRR (IV, R)	**0.69 [0.59, 0.81]**	**<0.001**	0%
Test for subgroup differences: $Chi^2 = 2.30$, df = 2 (P = 0.32), $I^2 = 13.0\%$						

2.1.6 Sensitivity analysis: risk of bias						
Incidence RTIs: some concerns RoB [7 22 23]	3	2767(3306)	IRR (IV, R)	**0.68 [0.56, 0.81]**	**<0.001**	0%
Incidence mild RTIs	*Not relevant: the three studies do not have high risk of bias*					
Incidence moderate RTIs	*and all three studies have some concerns*					
Incidence RTIs older adults	*Not relevant: only one study with some concerns*					
Incidence young adults: some concerns RoB [22 23]	2	2718 (2718)	IRR (IV, R)	**0.69 [0.59, 0.81]**	**<0.001**	0%

RTI: Respiratory tract infection, **HRV:** Human rhinovirus, **CI:** Confidence interval, **IRR:** Incidence rate ratio, **IRD:** Incidence rate difference, **IV:** Inverse variance method, **R:** Random effects model, **I^2:** % variation due to heterogeneity, *I^2 P value ≤ 0.05, **I^2 P value ≤ 0.01, ***I^2 P value ≤ 0.001

2.2 Prevention of RTIs caused by human rhinovirus (HRV) inoculation: short-term (2 – 36 hours) Zinc vs. placebo

Outcome	Studies	Participants	Statistical Method	Effect estimate [95% CI]	P value	I^2
2.2.1 Risk of clinical cold per person PrEP or PEP						
Incidence clinical cold [24-26]	4	221	RR (M-H, R)	0.96 [0.77, 1.21]	0.75	0%
2.2.2 Risk of laboratory confirmed HRV infection per person PrEP or PEP						
Incidence infection [24-26]	4	221	RR (M-H, R)	1.00 [0.91, 1.10]	0.93	0%
2.2.3 Risk of clinical cold per person PrEP						
Incidence clinical cold [24 26]	2	148	RR (M-H, R)	1.05 [0.69, 1.60]	0.82	0%
2.2.4 Risk of laboratory confirmed HRV infection per person PEP						
Incidence infection [24 26]	2	148	RR (M-H, R)	1.00 [0.85, 1.17]	1.00	10%
2.2.5 Risk of clinical cold per person PEP						
Incidence clinical cold [25]	2	73	RR (M-H, R)	0.92 [0.66, 1.29]	0.64	34%
2.2.6 Risk of laboratory confirmed HRV infection per person PrEP						
Incidence infection [25]	2	73	RR (M-H, R)	1.01 [0.89, 1.14]	0.90	0%
2.2.7 Sensitivity analysis: risk of bias						
Not relevant, the study outcomes do not have high risk of bias						

HRV: Human rhinovirus, **PrEP:** Pre-exposure prophylaxis, PEP**:** Post-exposure prophylaxis, **CI:** Confidence interval, **RR:** Risk ratio, **M-H:** Mantel-Haenszel method, **R:** Random effects model, **I^2:** % variation due to heterogeneity, *I^2 P value ≤ 0.05, **I^2 P value ≤ 0.001

3 META-ANALYSIS RESULTS: TREATMENT OF SYMPTOM SEVERITY

3.1 Symptom severity of mild to moderate RTI: Zinc vs. placebo

Outcome	Studies	Participants	Statistical Method	Effect estimate [95% CI]	P value	I^2
3.1.1 Symptom severity at the peak of infection: composite score on day 3 of common colds						
Mean day 3 score [26-30]	5	392	MD (IV, R)	**-1.20 [-1.74, -0.67]**	**<0.001**	0%
3.1.2 Symptom severity per day: average daily composite score						
Mean av. daily score [25 26 31]	3	195	SMD (IV, R)	-0.15 [-0.43, 0.13]	0.31	0%

3.1.3 Symptom severity at the peak of infection (day 3) according to type of infection						
Mean day 3 score community acquired [27-30]	4	323	MD (IV, R)	**-1.36 [-2.14, -0.58]**	**0.001**	0%
Mean day 3 score HRV inoculation [26]	1	69	MD (IV, R)	**-1.05 [-1.80, -0.30]**	**0.006**	n/a
Test for subgroup differences: Chi² = 0.33, df = 1 (P = 0.57), I² = 0%						
Mean av. daily score community acquired [31]	1	101	SMD (IV, R)	-0.31 [-0.71, 0.08]	0.12	n/a
Mean av. daily score HRV inoculation [25 26]	2	94	SMD (IV, R)	0.03 [-0.38, 0.44]	0.88	0%
Test for subgroup differences: Chi² = 1.43, df = 1 (P = 0.23), I² = 29.9%						
3.1.4 Symptom severity according to administration route						
Mean day 3 score sublingual lozenge [27-29]	3	163	MD (IV, R)	**-1.66 [-2.65, -0.66]**	**0.001**	0%
Mean day 3 score topical nasal [26 30]	2	229	MD (IV, R)	**-1.01 [-1.65, -0.37]**	**0.002**	0%
Test for subgroup differences: Chi² = 1.14, df = 1 (P = 0.3), I² = 12.2%						
Mean av. daily score sublingual lozenge [25 31]	2	126	SMD (IV, R)	-0.12 [-0.66, 0.42]	0.65	42%
Mean av. daily score topical nasal [26]	1	69	SMD (IV, R)	-0.06 [-0.53, 0.42]	0.81	n/a
Test for subgroup differences: Chi² = 0.03, df = 1 (P = 0.86), I² = 0%						
3.1.5 Sensitivity analyses: risk of bias						
Mean day 3 score: some concerns RoB [28-30]	3	258	MD (IV, R)	**-1.19 [-2.05, -0.33]**	**0.007**	0%
Mean av. daily score: some concerns RoB [25]	1	25	SMD (IV, R)	0.27 [-0.51, 1.06]	0.50	n/a

RTI: Respiratory tract infection, **HRV:** Human rhinovirus, **CI:** Confidence interval, **av.:** average, **MD:** Mean difference, **SMD:** Standardised mean difference **IV:** Inverse variance method, **R:** Random effects model, **I²:** % variation due to heterogeneity, *I^2 P value ≤ 0.05, **I^2 P value ≤ 0.01, ***I^2 P value ≤ 0.001

4 META-ANALYSIS RESULTS: DURATION OF SYMPTOMS

4.1 Duration of symptoms from community acquired RTIs: Zinc vs. placebo

Outcome	Studies	Participants	Statistical Method	Effect estimate [95% CI]	P value	I^2
4.1.1 Likelihood on any given day of symptomatic recovery from mild RTIs (up to 7 days)						
Symptomatic risk [a 27 29 30 32-38]	10	1023	HR (IV, R)	**1.83 [1.07, 3.13]**	**0.03**	**82%*** **
			Absolute effect [b]	**0.19 [0.02,0.38]**		
4.1.2 Mean duration (days) of symptoms from community acquired mild RTIs						
Mean duration (days) [27-31 34-40]	12	1180	MD (IV, R)	**-2.05 [-3.50, -0.59]**	**0.006**	**97%*** **

4.1.3 Duration according to administration route						
Symptomatic risk [a] sublingual lozenge [27 29 32 33 36-38]	7	572	HR (IV, R)	1.43 [0.85, 2.40]	0.18	**69%****
Symptomatic risk [a] topical nasal [30 34 35]	3	451	HR (IV, R)	3.38 [0.79, 14.45]	0.10	**93%*****
Test for subgroup differences: $Chi^2 = 1.20$, df = 1 (P = 0.27), $I^2 = 17.0\%$						
Mean duration (days) sublingual lozenge [27-29 31 36-40]	9	729	MD (IV, R)	**-1.84 [-3.08, -0.59]**	**0.004**	**90%****
Mean duration (days) topical nasal [30 34 35]	3	451	MD (IV, R)	-2.88 [-6.61, 0.85]	0.13	**97%*****
Test for subgroup differences: $Chi^2 = 0.27$, df = 1 (P = 0.60), $I^2 = 0\%$						
4.1.4 Duration according to zinc lozenge dose						
Symptomatic risk [a] < 50mg daily [38]	1	130	HR (IV, R)	0.83 [0.44, 1.55]	0.55	n/a
Symptomatic risk [a] 50 – 200mg daily [29 33 36 37]	4	334	HR (IV, R)	1.58 [0.71, 3.53]	0.26	**80%****
Symptomatic risk [a] > 200mg daily [27 32]	2	98	HR (IV, R)	1.74 [0.78, 3.84]	0.17	21%
Test for subgroup differences: $Chi^2 = 2.64$, df = 2 (P = 0.27), $I^2 = 24.3\%$						
Mean duration (days) < 50mg daily [38]	1	130	MD (IV, R)	0.18 [-0.17, 0.52]	0.31	n/a
Mean duration (days) 50 – 200mg daily [28 29 31 36 37 39]	6	471	MD (IV, R)	**-0.92 [-1.66, -0.18]**	**0.01**	**93%****
Mean duration (days) > 200mg daily [27 40]	2	128	MD (IV, R)	**-0.85[-1.65, -0.08]**	**0.03**	**77%****
Test for subgroup differences: **$Chi^2 = 10.85$, df = 2 (P = 0.004), $I^2 = 81.6\%$**						
4.1.5 Duration according to zinc salt						
Symptomatic risk [a] zinc acetate [29 33]	2	135	HR (IV, R)	1.73 [0.23, 13.05]	0.56	**92%*****
Symptomatic risk [a] zinc gluconate [27 33-38]	7	782	HR (IV, R)	1.86 [0.92, 3.75]	0.08	**86%*****
Symptomatic risk [a] zinc orotate & gluconate [32]	1	33	HR (IV, R)	1.02 [0.31, 3.37]	0.97	n/a
Symptomatic risk [a] zinc sulfate [30]	1	160	HR (IV, R)	1.16 [0.68, 1.97]	0.58	n/a
Test for subgroup differences: $Chi^2 = 1.39$, df = 3 (P = 0.71), $I^2 = 0\%$						
Mean duration (days) zinc acetate [28 29 31 39]	4	262	MD (IV, R)	**-2.44 [-3.77, -1.11]**	**<0.001**	**77%****
Mean duration (days) zinc gluconate [27 34-38 40]	7	758	MD (IV, R)	**-2.45 [-4.68, -0.22]**	**0.03**	**98%*****
Mean duration (days) zinc sulfate [30]	1	160	MD (IV, R)	-0.22 [-1.07, 0.63]	0.61	n/a

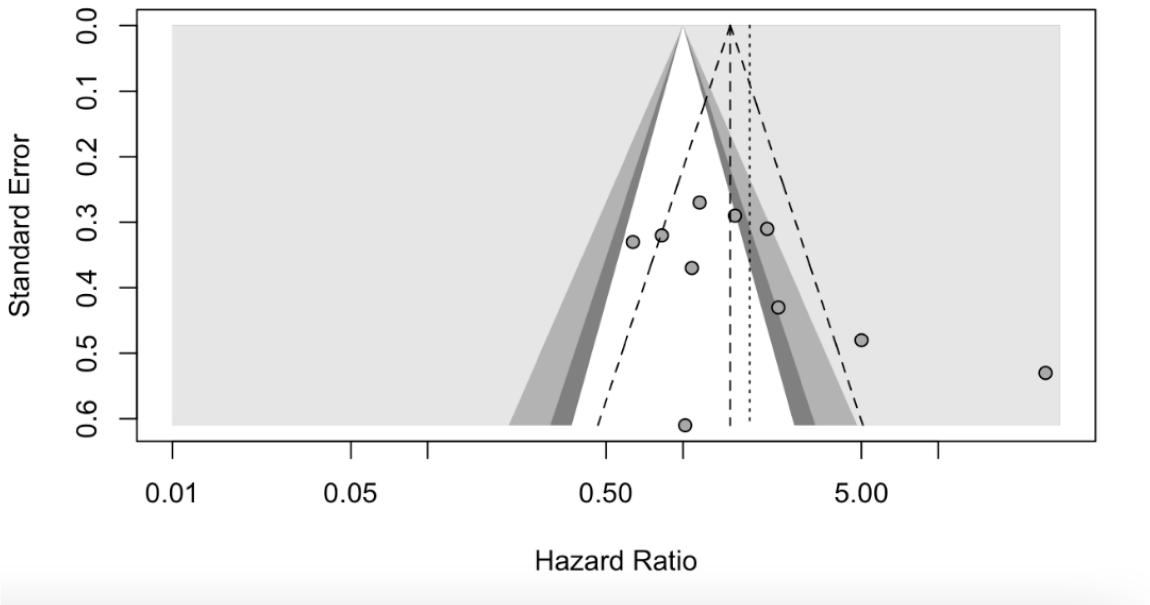
Test for subgroup differences: **Chi² = 9.41, df = 2 (P = 0.009), I² = 78.7%**						
4.1.6 Duration according to days symptomatic prior to study enrolment – *post hoc*						
Symptomatic risk: [a] ≤ 3 days symptomatic when started [27 32]	2	98	HR (IV, R)	1.74 [0.78, 3.84]	0.26	21%
Symptomatic risk: [a] ≤ 2 days symptomatic when started [30]	1	160	HR (IV, R)	1.16 [0.68, 1.97]	0.58	n/a
Symptomatic risk: [a] ≤ 1 day symptomatic when started [29 33-36 38]	6	655	HR (IV, R)	2.33 [0.97, 5.55]	0.06	**89%*** **
Symptomatic risk: [a] timing unknown [37]	1	110	HR (IV, R)	1.08 [0.52, 2.24]	0.83	n/a
Test for subgroup differences: Chi² = 2.54, df = 3 (P = 0.47), I² = 0%						
Mean duration ≤ 3 days symptomatic when started [27]	1	65	MD (IV, R)	**-3.62 [-5.07, -2.17]**	**<0.001**	**n/a**
Mean duration ≤ 2 days symptomatic when started [30 35 39 40]	4	364	MD (IV, R)	-0.79 [-1.99, 0.41]	0.20	**81%****
Mean duration ≤ 1 day symptomatic when started [28 29 34 36 38]	5	540	MD (IV, R)	**-3.39 [-6.02, -0.76]**	**0.01**	**98%*** **
Mean duration timing started unknown [31 37]	2	211	MD (IV, R)	-0.98 [-2.40, 0.44]	0.18	63%
Test for subgroup differences: **Chi² = 11.48, df = 3 (P = 0.009), I² = 73.9%**						
4.1.7 Duration according to definition of recovery – *post hoc*						
Symptomatic risk: [a] recovery is asymptomatic [27 32 34 35 37 38]	6	629	HR (IV, R)	2.04 [0.88, 4.73]	0.10	**86%*** **
Symptomatic risk: [a] recovery is 48 hrs asymptomatic [30]	1	160	HR (IV, R)	1.16 [0.68, 1.97]	0.58	n/a
Symptomatic risk: [a] recovery is symptom score ≤ 1 [29 36]	2	147	HR (IV, R)	**3.02 [1.33, 6.85]**	**0.008**	55%
Symptomatic risk: [a] recovery is self-determined [33]	1	87	HR (IV, R)	0.64 [0.33, 1.22]	0.17	n/a
Test for subgroup differences: **Chi² = 10.00, df = 3 (P = 0.02), I² = 70.0%**						
Mean duration recovery is asymptomatic [27 29 31 34 35 37 38 40]	8	808	MD (IV, R)	**-2.33 [-4.28, -0.38]**	**0.01**	**97%*** **
Mean duration recovery 48 hrs asymptomatic [30]	1	160	MD (IV, R)	-0.22 [-1.07, 0.63]	0.61	n/a
Mean duration recovery symptom score ≤ 1 [28 36 39]	3	212	MD (IV, R)	-2.25 [-4.63, 0.13]	0.06	81%**
Test for subgroup differences: Chi² = 5.55, df = 2 (P = 0.06), I² = 64.0%						
4.1.8 Sensitivity analysis: Tierney et al. method 11 to estimate hazard ratio [a]						
Symptomatic risk [27 29 30 32-38]	10	1023	HR (IV, R)	**1.99 [1.06, 3.73]**	**0.03**	**93%*** **
4.1.9 Sensitivity analyses: addition of Turner 2000[19] with an active control (3 zinc arms combined)						
Symptomatic risk [a 19 27 29 30 32-38]	12	1570	HR (IV, R)	**1.61 [1.07, 2.43]**	**0.02**	**80%*** **
Mean days duration [c 19 27-31 34-40]	13	1459	MD (IV, R)	**-1.89 [-3.29, -0.50]**	**0.008**	**97%*** **

4.1.10 Sensitivity analyses: risk of bias						
Symptomatic risk:[a] low RoB [29 30]	2	208	HR (IV, R)	2.28 [0.55, 9.52]	0.26	**86%****
Symptomatic risk:[a] some concerns or low RoB[29 30 33-36 38]	6	815	HR (IV, R)	**2.06 [1.01, 4.22]**	**0.05**	**87%*****
Mean duration: low RoB [28-30]	2	108	MD (IV, R)	-1.90 [-5.21, 1.41]	0.26	**96%*****
Mean duration: some concerns or low RoB [28-30 34-36 39 40]	9	942	MD (IV, R)	**-2.44 [-4.12, -0.76]**	**0.004**	**98%*****
4.1.11 Sensitivity analyses: removal of statistical outliers						
Symptomatic risk (publication bias) [a 27 29 30 32 33 35-38]	9	810	HR (IV, R)	1.39 [0.96, 2.02]	0.08	**60%****
Symptomatic risk (heterogeneity) [a 27 30 32 35-38]	7	675	HR (IV, R)	**1.37 [1.03, 1.81]**	**0.03**	19%
Mean days duration (heterogeneity) [27-31 35-38 40]	10	904	MD (IV, R)	**-1.86 [-2.74, -0.98]**	**<0.001**	**90%*****

RTI: Respiratory tract infection, **CI:** Confidence interval, **HR:** Hazard ratio, **MD:** Mean difference, **IV:** Inverse variance method, **R:** Random effects model, **I^2:** % variation due to heterogeneity, *I^2 P value ≤ 0.05, **I^2 P value ≤ 0.01, ***I^2 P value ≤ 0.001
[a] Tierney method 10 was selected in preference to method 11 based on greater precision and HR favouring the null hypothesis[15]
[b] Absolute effect calculated in GRADEpro GDT[6] probability of remaining symptomatic on day-7 was set *post hoc* at 33%.[21]
[c] The means and SD for the three zinc arms were combined in RevMan 5.4

4.1.12 Assessment of publication bias

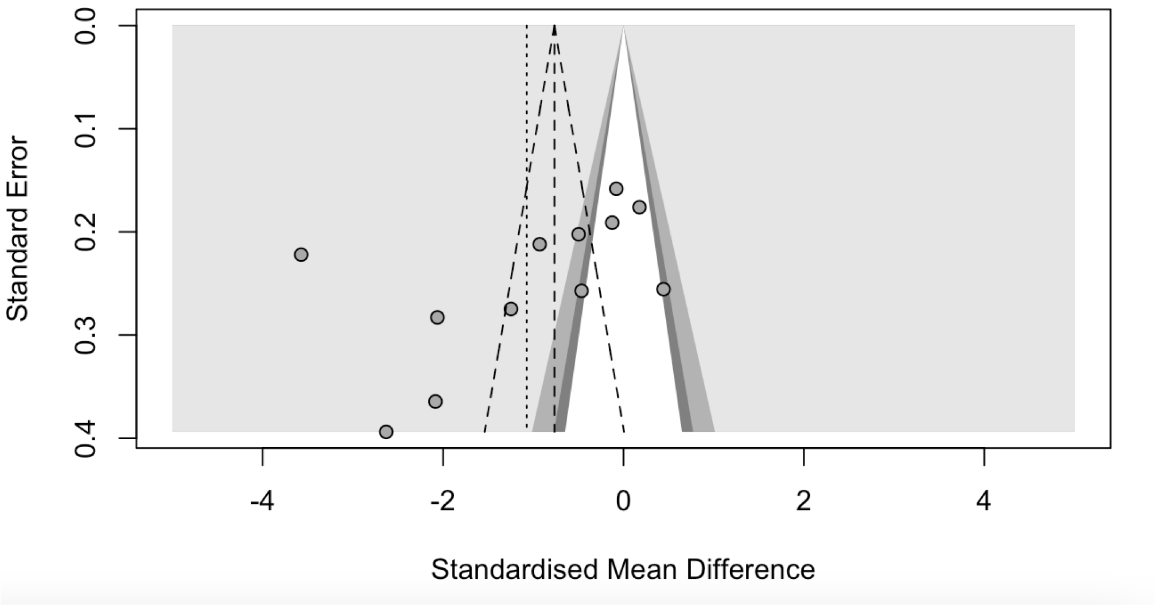
Figure 1. Funnel plot: Zinc vs. placebo on duration of community acquired RTIs, likelihood of symptomatic recovery



Visual inspection of the funnel plot is suggestive of asymmetry. However, the outlying study with the largest effect size, also had the largest sample size (n=213).[34] Heterogeneity can exacerbate funnel plot asymmetry.[16] Removal of this outlier substantially reduced this asymmetry and reduced statistical

heterogeneity from I^2=82% to 60% (Table 4.1.11. symptomatic risk, publication bias). Overall, small study bias is not strongly suspected.

Figure 2. Funnel plot: Zinc vs. placebo on duration of community acquired RTIs, mean number of days symptomatic



Visual inspection of the funnel plot shows asymmetry that is suggestive of small study bias. However, Egger's regression was not significant (p = 0.54). Overall, small study bias is not strongly suspected.

4.2 Duration of symptoms from community acquired and HRV inoculated RTI: Zinc vs. active control

Outcome	Studies	Participants	Statistical Method	Effect estimate [95% CI]	P value	I^2
4.2.1 Likelihood of recovery from mild RTIs on any given day for up to 7 days						
Asymptomatic risk [a 19]	2	547	HR (IV, R)	1.06 [0.79, 1.41]	0.71	0%
			Absolute Risk [d]	0.02 [-0.03, 0.07]		
4.2.2 Mean duration of symptoms (days) from community acquired mild RTI						
Mean duration (days) [19]	1	279	MD (IV, R)	-0.16 [-1.22, 0.90]	0.80	n/a
4.2.3 Duration according to type of infection						
Asymptomatic risk community acquired [19]	1	275	HR (IV, R)	1.03 [0.64, 1.65]	0.90	n/a
Asymptomatic risk HRV inoculation [19]	1	272	HR (IV, R)	1.07 [0.74, 1.56]	0.71	n/a
Test for subgroup differences: Chi^2 = 0.02, df = 1 (P = 0.90), I^2 = 0%						
4.2.4 Duration according to zinc dose						
Asymptomatic risk < 50mg daily [19]	2 [c]	276 [c]	HR (IV, R)	0.93 [0.69, 1.25]	0.62	0%

Asymptomatic risk 50 – 200mg daily [19]	4 [c]	551 [c]	HR (IV, R)	1.14 [0.93, 1.41]	0.25	0%
Test for subgroup differences: Chi² = 1.30, df = 1 (P = 0.25), I² = 22.9%						
Mean duration (days) < 50mg daily [19]	1	97	MD (IV, R)	0.35 [-1.49, 2.19]	0.71	n/a
Mean duration (days) 50 – 200mg daily [19]	2	182	MD (IV, R)	-0.41 [-1.70, 0.89]	0.54	0%
Test for subgroup differences: Chi² = 0.44, df = 1 (P = 0.51), I² = 0%						
4.2.5 Duration according to zinc salt						
Asymptomatic risk zinc acetate [19]	4 [c]	552 [c]	HR (IV, R)	1.00 [0.81, 1.24]	0.96	0%
Asymptomatic risk zinc gluconate [19]	2 [c]	275 [c]	HR (IV, R)	1.19 [0.90, 1.59]	0.22	0%
Test for subgroup differences: Chi² = 0.91, df = 1 (P = 0.34), I² = 0%						
Mean duration (days) zinc acetate [19]	2	282	MD (IV, R)	0.11 [-1.20, 1.42]	0.71	0%
Mean duration (days) zinc gluconate [19]	1	139	MD (IV, R)	-0.66 [-2.46, 1.14]	0.47	n/a
Test for subgroup differences: Chi² = 0. 46, df = 1 (P = 0.50), I² = 0%						
4.2.6 Sensitivity analysis: Tierney et al. method 11 to estimate hazard ratio [a]						
Symptomatic risk [19]	2	547	HR (IV, R)	1.10 [0.89, 1.38]	0.38	0%
4.2.7 Sensitivity analysis: no. participants in placebo group divided into 3-arms						
Mean duration (days) [19]	3	279	MD (IV, R)	-0.18 [-0.93, 0.56]	0.63	0%
4.2.8 Sensitivity analysis: unit of analysis error ignored						
Asymptomatic risk [a 19]	6 [c]	827 [c]	HR (IV, R)	1.07 [0.90, 1.27]	0.45	0%
Mean duration (days) [19]	3	421	MD (IV, R)	-0.18 [-0.93, 0.56]	0.63	0%
4.2.9 Sensitivity analysis: risk of bias						
Not relevant, the study outcomes do not have high risk of bias, all studies and study arms have some concerns						

RTI: Respiratory tract infection, **HRV:** Human rhinovirus, **CI:** Confidence interval, **HR:** Hazard ratio, **MD:** Mean difference, **IV:** Inverse variance method, **R:** Random effects model, **I²:** % variation due to heterogeneity, *I^2 P value ≤ 0.05, **I^2 P value ≤ 0.01, ***I^2 P value ≤ 0.001 [a] Tierney method 10 was selected in preference to method 11 based on greater precision and HR favoured the null hypothesis; [b] The means and SD for the three zinc arms were combined in RevMan 5.4; [c] NOTE: unit of analysis error as events in placebo arms are counted three times, however, sensitivity analysis (4.2.7) suggests only small, insignificant conflation from unaddressed correlation. [d] Absolute effect calculated in GRADEpro GDT[6] probability of remaining symptomatic on day-7 was set *post hoc* at 33%.[21]

5 META-ANALYSIS RESULTS: ADVERSE EVENTS

5.1 Tolerance and safety studies: Zinc lozenges vs. placebo

Outcome	Studies	Participants	Statistical Method	Effect estimate [95% CI]	P value	I^2
5.1.1 Risk of any adverse events (AE) per asymptomatic participant						
Incidence of AE [41]	1	66	RR	1.33 [0.32, 5.50]	0.69	n/a

RTI: Respiratory tract infection, **CI:** Confidence interval, **RR:** Risk ratio, **I^2:** % variation due to heterogeneity, *I^2 P value ≤ 0.05, **I^2 P value ≤ 0.01, ***I^2 P value ≤ 0.001

5.2 Adverse Effects for prevention of RTI: Zinc vs. placebo

Outcome	Studies	Participants (person-months)	Statistical Method	Effect estimate [95% CI]	P value	I^2
5.2.1 Risk of any adverse events (AE) per person-month						
Incidence of any AE [8 22 23]	3	2758 (2,998)	IRR (IV, R)	1.63 [0.81, 3.31]	0. 17	**62%**
			IRD (IV, R)	0.02 [-0.02, 0.05]	0.44	**85%*** **
5.2.2 Risk of nausea or stomach-ache from 15mg zinc daily in oral capsules						
Incidence of nausea or stomach-ache [8]	1	40 (280)	IRR	0.17 [0.01, 3.51]	0.25	n/a
5.2.3 Risk of nasal irritation/soreness from nasal gels/sprays						
Incidence of nasal irritation/soreness [22 23]	2	2718 (2718)	IRR (IV, R)	1.80 [0.92, 3.53]	0.09	**74%***
5.2.4 Risk of copper deficiency in adults 55-87 years from 45mg zinc daily for 12-months						
Serum copper µg/dL [7]	1	49 (98)	MD	-4.70 [-38.15, 28.75]	0.78	n/a
5.2.5 Sensitivity analysis: risk of bias						
Incidence of any AE [22 23]	2	2918 (2718)	IRR (IV, R)	1.80 [0.92, 3.53]	0.09	**74%***
5.2.6 Sensitivity analysis: RCT with zero events in zinc arm excluded						
Incidence of any AE [22 23]	2	2718 (2718)	IRR (IV, R)	1.80 [0.92, 3.53]	0.09	**74%***
5.2.7 Sensitivity analysis: RCT with zero events in zinc arm replaced with 1 event						
Incidence of any AE [8 22 23]	3	2758 (2,998)	IRR (IV, R)	**1.81 [1.08, 3.01]**	**0.02**	**45%**

RTI: Respiratory tract infection, **AE:** Adverse effects, **CI:** Confidence interval, **IRR:** Incidence rate ratio, **RD:** Incidence rate difference, **MD:** Mean difference, **IV:** Inverse variance method, **R:** Random effects model, **I^2:** % variation due to heterogeneity, ***I^2** P value ≤ 0.05, ****I^2** P value ≤ 0.01, *****I^2** P value ≤ 0.001, **µg/dL:** micrograms per decilitre of blood

5.3 Adverse events for treatment of RTI: Zinc vs. placebo

Outcome	Studies	Participants	Statistical Method	Effect estimate [95% CI]	P value	I^2
5.3.1 Risk of any adverse events (AE)						
Incidence of any AE [25-27 30 32-36 38 40]	11	1102	RR (M-H, R)	**1.41 [1.17, 1.69]**	**<0.001**	36%

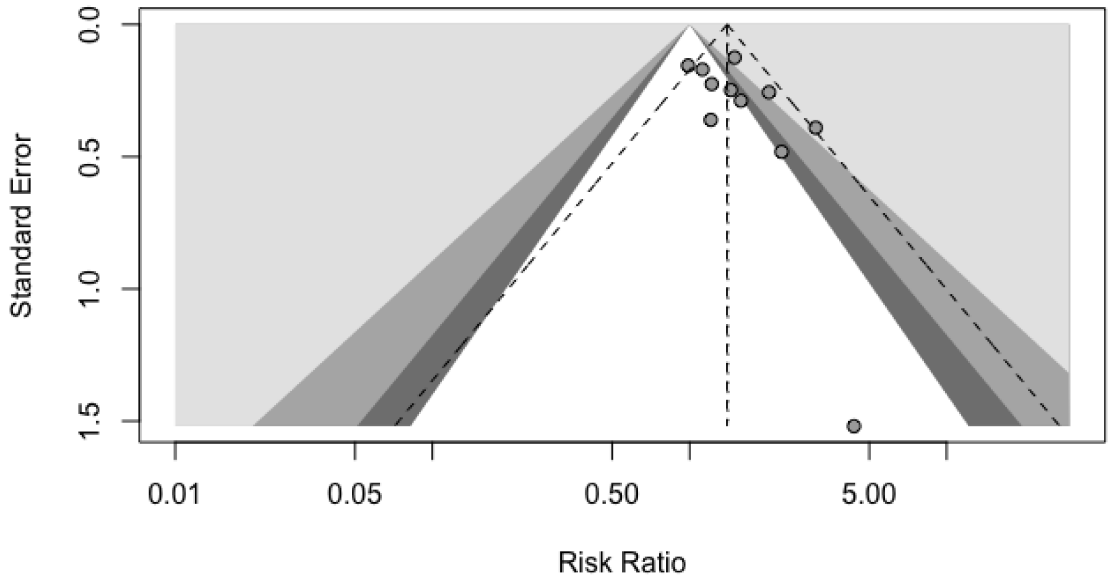
			RD (M-H, R)	**0.14 [0.09, 0.20]**	**<0.001**	**49%***
5.3.2 Risk of nausea or stomach-ache						
Incidence of nausea or stomach-ache [25 27-29 32-37 39 40]	12	1015	RR (M-H, R)	**1.46 [1.03, 2.06]**	**0.04**	9%
5.3.3 Risk of mouth irritation or soreness from sublingual lozenges						
Incidence of mouth irritation or soreness [25 27-29 32 37 40]	7	505	RR (M-H, R)	**1.55 [1.05, 2.29]**	**0.03**	1%
5.3.4 Risk of taste aversion from sublingual lozenges						
Incidence of taste aversion [27-29 32 36-39]	9	719	RR (M-H, R)	**2.11 [1.47, 3.04]**	**<0.001**	24%
5.3.5 Risk of nasal irritation from nasal gels/sprays						
Incidence of nasal irritation [26 30 35]	3	328	RR (M-H, R)	1.22 [0.72, 2.05]	0.45	0%
5.3.6 Risk of any AE according to administration route						
Sublingual lozenges [25 27 33 36 38 40]	6	514	RR (M-H, R)	**1.64 [1.36, 1.99]**	**<0.001**	3%
Nasal spray/gel [26 30 34 35]	4	541	RR (M-H, R)	1.12 [0.92, 1.36]	0.28	1%
Lozenge & nasal spray [32]	1	47	RR (M-H, R)	1.21 [0.60, 2.46]	0.60	n/a
Test for subgroup differences: **Chi² = 7.66, df = 2 (P = 0.02), I² = 73.9%**						
5.3.7 Risk of nausea or stomach-ache according to administration route						
Sublingual lozenges [25 28 29 33 35 36 39 40 42]	11	830	RR (M-H, R)	**1.48 [1.00, 2.18]**	**0.05**	16%
Topical nasal gel/spray [30 34]	2	372	RR (M-H, R)	1.44 [0.24, 8.72]	0.69	0%
Lozenge & nasal spray [32]	1	47	RR (M-H, R)	1.72 [0.59, 5.02]	0.32	n/a
Test for subgroup differences: Chi² = 0.07, df = 2 (P = 0.97), I² = 0%						
5.3.8 Risk of any AE according to daily zinc dose						
Zinc <50 mg/day [26 30 34 35 38]	5	671	RR (M-H, R)	1.17 [0.96, 1.43]	0.1	9%
Zinc 50-200 mg/day [25 33 36 40]	4	303	RR (M-H, R)	**1.57 [1.29, 1.92]**	**<0.001**	0%
Zinc >200 mg/day [27 32]	2	128	RR (M-H, R)	1.91 [0.75, 4.91]	0.2	69%
Test for subgroup differences: Chi² = 4.69, df = 2 (P = 0.10), I² = 57.4%						
5.3.9 Risk of nausea or stomach-ache according to daily zinc dose						
Zinc <50 mg/day [30 34]	2	372	RR (M-H, R)	1.44 [0.24, 8.72]	0.69	0%
Zinc 50-200 mg/day [25 29 33 36 37 39 40]	9	671	RR (M-H, R)	1.41 [0.96, 2.06]	0.08	13%
Zinc >200 mg/day [27 32]	2	136	RR (M-H, R)	3.48 [0.42, 28.58]	0.25	55%
Test for subgroup differences: Chi² = 0.69, df = 2 (P = 0.71), I² = 0%						
5.3.10 Risk of mouth irritation/soreness from sublingual lozenges according to daily zinc dose						
Zinc 50-200mg/day [25 28 29 37 40]	5	379	RR (M-H, R)	**1.51 [1.01, 2.27]**	**0.05**	6%
Zinc >200mg/day [27 32]	2	128	RR (M-H, R)	3.02 [0.35, 25.79]	0.31	52%
Test for subgroup differences: Chi² = 0.39, df = 1 (P = 0.53), I² = 0%						
5.3.11 Risk of taste aversion from sublingual lozenges according to daily zinc dose						
Zinc < 50mg/day [38]	1	130	RR (M-H, R)	7.90 [0.42, 150.01]	0.17	n/a
Zinc 50-200mg/day [25 28 29 36 37 39]	6	461	RR (M-H, R)	**2.13 [1.34, 3.40]**	**0.001**	47%

Zinc >200mg/day [27 32]	2	128	RR (M-H, R)	2.12 [0.81, 5.55]	0.13	0%
Test for subgroup differences: $Chi^2 = 0.75$, df = 2, (P = 0.69), $I^2 = 0\%$						
5.3.12 Risk of any AE according to zinc salt						
Zinc acetate [33]	1	88	RR (M-H, R)	**2.04 [1.23, 3.37]**	**0.006**	n/a
Zinc gluconate [25-27 34-36 38 40]	8	808	RR (M-H, R)	**1.45 [1.21, 1.74]**	**<0.001**	16%
Zinc orotate & gluconate [32]	1	49	RR (M-H, R)	1.32 [0.64, 2.71]	0.45	n/a
Zinc sulfate [30]	1	159	RR (M-H, R)	0.99 [0.73, 1.34]	0.93	n/a
Test for subgroup differences: $Chi^2 = 7.11$, df = 3 (P = 0.07), $I^2 = 57.8\%$						
5.3.13 Risk of nausea or stomach-ache according to zinc salt						
Zinc acetate [28 29 33 39]	4	249	RR (M-H, R)	1.53 [0.47, 4.95]	0.48	34%
Zinc gluconate [25 27 34 36 37 40]	7	716	RR (M-H, R)	**1.47 [1.02, 2.10]**	**0.04**	0%
Zinc orotate & gluconate [32]	1	55	RR (M-H, R)	14.48 [0.87, 241.82]	0.06	n/a
Zinc sulfate [30]	1	179	RR (M-H, R)	1.93 [0.18, 20.81]	0.59	n/a
Test for subgroup differences: $Chi^2 = 2.54$, df = 3 (P = 0.47), $I^2 = 0\%$						
5.3.14 Risk of mouth irritation or soreness from sublingual lozenges according to zinc salt						
Zinc acetate [28 29]	2	98	RR (M-H, R)	1.56 [0.42, 5.77]	0.51	28%
Zinc gluconate [25 27 37 40]	4	360	RR (M-H, R)	1.62 [0.88, 2.97]	0.12	38%
Zinc orotate & gluconate [32]	1	47	RR (M-H, R)	1.47 [0.40, 5.44[	0.57	n/a
Test for subgroup differences: $Chi^2 = 0.02$, df = 2 (P = 0.99), $I^2 = 0\%$						
5.3.15 Risk of taste aversion from sublingual lozenges according to zinc salt						
Zinc acetate [28 29 39]	3	156	RR (M-H, R)	1.79 [0.81, 3.94]	0.15	58%
Zinc gluconate [25 27 36-38]	5	516	RR (M-H, R)	**2.76 [1.88, 4.07]**	**<0.001**	0%
Zinc orotate & gluconate [32]	1	47	RR (M-H, R)	1.76 [0.50, 6.22]	0.38	n/a
Test for subgroup differences: $Chi^2 = 1.24$, df = 2 (P = 0.54), $I^2 = 0\%$						
5.3.16 Sensitivity analysis: risk of bias						
Incidence of any AE: some or low RoB [25 26 30 33-36 40]	10	974	RR (M-H, R)	**1.35 [1.14, 1.60]**	**<0.001**	26%
Incidence of any AE: low RoB only [25 30 33 36]	4	389	RR (M-H, R)	**1.41 [0.99, 2.02]**	**0.05**	62%
Incidence of nausea or stomach-ache: some or low RoB[25 33 34 36 40]	6	549	RR (M-H, R)	1.22 [0.82, 1.83]	0.33	0%
Incidence of mouth irritation or soreness: some or low RoB[25 40]	2	105	RR (M-H, R)	1.24 [0.74, 2.08]	0. 35	0%
Incidence of taste aversion: some or low RoB[25 38 40]	3	261	RR (M-H, R)	**2.70 [1.77, 4.12]**	**<0.001**	0%
Incidence of nasal irritation	*Not relevant, the study outcomes do not have high risk of bias, all studies have some concerns*					

RTI: Respiratory tract infection, **AE:** Adverse effects, **CI:** Confidence interval, **RR:** Risk ratio, **MD:** Mean difference, **IV:** Inverse variance method, **R:** Random effects model, **I^2:** % variation due to heterogeneity, *I^2 P value ≤ 0.05, **I^2 P value ≤ 0.01, ***I^2 P value ≤ 0.001, µg/dL: micrograms per decilitre of blood

5.3.17 Assessment of publication bias

Figure 3. Funnel plot: Any adverse events for treatment of RTI: Zinc vs. placebo



Visual inspection of the funnel plot showed some asymmetry. However, the asymmetry is in favour of lower risk for placebo controls, which is the opposite of what is expected if there was publication bias, and the Harbord score was not significant (p = 0.073). Overall, small study bias is not strongly suspected.

5.4 Adverse effects for treatment of RTI: Zinc vs. active controls

Outcome	Studies	Participants	Statistical Method	Effect estimate [95% CI]	P value	I^2
5.4.1 Risk of any adverse effects (AE)						
Incidence of any AE [19 43]		703	RR (M-H, R)	1.12 [0.76, 1.65]	0.57	0%
		703	RD (M-H, R)	0.02 [-0.03, 0.07]	0.40	0%
5.4.2 Risk of nasal burning from nasal gels/sprays containing zinc vs. naphazoline						
Incidence of nasal burning [43]	1	151	RR (M-H, R)	0.34 [0.01 to 8.16]	0.50	n/a
5.4.3 Risk of taste aversion from sublingual lozenges containing zinc vs. quinine						
Incidence of taste aversion [19]	1	279	RR (M-H, R)	0.94 [0.31, 2.85]	0.91	n/a
5.4.4 Sensitivity analysis: risk of bias						
Incidence of taste aversion: low RoB [19]	2		RR (M-H, R)	1.17 [0.71, 1.92]	0.35	26%

RTI: Respiratory tract infection, **AE:** Adverse effects, **GI:** Gastrointestinal, **CI:** Confidence interval, **RR:** Risk ratio, **IV:** Inverse variance model **M-H:** Mantel-Haenszel method, **R:** Random effects model, **I^2:** % variation due to heterogeneity, *I^2 P value ≤ 0.05, **I^2 P value ≤ 0.01, ***I^2 P value ≤ 0.001

6 OTHER CALCULATIONS

6.1 Daily dose estimates for Zinc Gluconate Nasal Sprays

Molecular Weight Zinc Gluconate: 455.7 g/mol [44]

Turner 2001 [26]
Preparation: Zicam (No manufacturer cited – currently produced by Matrixx Initiatives, Inc Bridgewater, NJ)
Formula: 33 m*M* zinc gluconate
Zinc Gluconate in Formula = (455.7 x 33)/1000g = 15.04g mixed into emulsion per litre of solution
Zinc Gluconate per Dose: Single nasal spray of 120 µL per nostril (ie two sprays per dose)
1 spray = 15.04g x (120µL/1000mL) = 15.04g x (0.12mL/1000mL)
= 15.04g x 0.00012 = 0.001805g = 1.8mg zinc gluconate
2 sprays per dose = 3.6mg zinc gluconate
Zinc as a percentage of zinc gluconate = 14.35% elemental Zinc by weight
Zinc per Nasal Spray:
1 spray = 1.8mg zinc gluconate = 0.26mg elemental zinc per spray
2 sprays = 0.52mg elemental zinc in a dose
Five doses per day = 2.6mg elemental zinc daily dose

Wei 2009 [22] **and Zhang 2009** [23]
Preparation: Shandong Tianshun Pharmaceutical Ltd.
Zinc Gluconate per Dose: This manufacture only has two specifications of zinc nasal spray:

- 15g: 0.3g (Each bottle contains 140 sprays, and each spray contains 2.0mg of zinc gluconate nasal spray)
- 10g: 0.2g (Each bottle contains 90 sprays, and each spray contains 2.0mg of zinc gluconate nasal spray)

Zinc as a percentage of zinc gluconate = 14.35% elemental Zinc by weight
Zinc per Nasal Spray:
1 spray = 2.0mg zinc gluconate = 0.287mg elemental zinc per spray
2 sprays = 0.57mg elemental zinc in a dose
Two doses per day = 1.15mg elemental zinc daily dose

6.2 Day-3 symptom severity score transformations

	Zinc mean	Zinc SD	Placebo mean	Placebo SD	No. items (score)	Score Range (Range ratio)
*Belongia 2001 [30]	4.5	4.1	5.4	4.0	8 (0 to 3)	24 (n/a)
Eby 1984 [27]	2.79	3.65	5.5	5.40	10 (0 to 3)	30 (0.80)
Prasad 2000 [29]	5.3	4.35	7.2	3.31	11 (0 to 4)	33 (0.73)
Prasad 2008 [28]	5.2	4.55	7.1	3.3	10 (0 to 3)	30 (0.80)
Turner 2001 [26]	2.1	0.51	3.5	3.04	8 (0 to 4)	32 (0.75)
TRANSFORMED **						
Eby 1984 [27]	2.23	2.92	4.40	4.32		
Prasad 2000 [29]	3.85	3.16	5.24	2.41		
Prasad 2008 [28]	4.16	3.64	5.68	2.64		
Turner 2001 [26]	1.58	0.38	2.63	2.28		

SD: Standard deviation; *Reference scale: Jackson cold scale, modified; **Transformation according to Thorlund et al. [10]

6.3 Day-3 symptom severity score minimally important difference

The minimally important difference (MID) in symptom severity for mild RTIs on day-3 was set at 1 point. This was the half-way mark between two proposed MIDs. Turner et al.[19] proposed a 10% improvement for mild RTIs. Based on the pooled mean scores for the control arms on day-3, a MID would be 0.5 points. Norman et al.[20] proposed that for populations with at least moderately impaired quality of life scores, the MID is approximately half the pooled standard deviation (SD) from the control arms, which for day-3 symptom severity would be 1.5 points.

Lower MID calculation: 10% improvement in the average mean placebo score[19]
Lower MID = (4.0 + 4.4 + 5.2 + 5.7 + 2.6)/5 x 10% = 0.47

Upper MID calculation: half the pooled standard deviation of zinc and placebo means[20]
Upper MID = √(2922.68/315) = 1.52

$$Pooled\ SD = \sqrt{\frac{(n_1 - 1) \times {SD_1}^2 + (n_2 - 1) \times {SD_2}^2}{n_1 + n_2 - 2}}$$

6.4 Survival curve data extraction for days symptomatic

	Eby 1984 Fig.1 [27]				Mossad 1996 Fig 1. [36]			
Day	Zinc %	no.	Control %	no.	Zinc %	no.	Control %	no.
0	100	37	100	28	100	49	100	50
1	78.4	29	100	28	91.8	45	100	50
2	62.2	23	92.9	26	81.6	40	92	46
3	54.1	20	82.1	23	69.4	34	86	43
4	35.1	13	78.6	22	53.1	26	76	38
5	29.7	11	71.4	20	44.9	22	72	36
6	18.9	7	67.9	19	32.7	16	62	31
7	13.5	5	53.6	15	18.4	9	56	28
	Eby 2006 Fig 1. [32]				Mossad 2003 Fig.2 [35]			
Day	Zinc %	no.	Control %	no.	Zinc %	no.	Control %	no.
0	100	16	100	17	100	40	100	38
1	87.5	14	100	17	92.5	37	97.4	37
2	81.3	13	94.1	16	82.5	33	94.7	36
3	68.8	11	76.5	13	67.5	27	86.8	33
4	56.3	9	76.5	13	50	20	71.1	27
5	43.8	7	64.7	11	35	14	52.6	20
6	43.8	7	58.8	10	20	8	39.5	15
7	37.5	6	47.1	8	15	6	26.3	11
	Hemilla 2020 [33]				Prasad 2000 Fig 1. [29]			
Day	Zinc %	no.	Control %	no.	Zinc %	no.	Control %	no.
0	100	45	100	42	100	25	100	23
1	100	45	100	42	100	25	100	23
2	88.9	40	90.5	38	92	23	100	23
3	80.0	36	76.2	32	64	16	100	23
4	68.9	31	64.3	27	48	12	100	23
5	60.0	27	42.9	18	28	7	91.3	21
6	57.8	26	26.2	11	12	3	82.6	19
7	37.8	17	26.2	11	0.01	0	65.2	15

	Hirt 2000 1 Fig.1 [34]				Smith 1989 Fig 1. [37]			
Day	Zinc %	no.	Control %	no.	Zinc %	no.	Control %	no.
0	100	108	100	105	100	57	100	53
1	100	108	100	105	100	57	98.1	52
2	48.1	52	100	105	100	57	96.2	51
3	30.5	33	98.1	103	96.4	55	92.5	49
4	18.5	2	98.1	103	86	49	79.2	42
5	0.01	0	95.2	100	77.2	44	67.8	36
6	0.001	0	90.5	95	57.9	33	62.3	33
7	0.0001	0	66.6	70	42.1	24	54.7	29
	Turner 2000 (A-1) Fig.1A [19]				Turner 2000 (B-1) Fig.1B [19]			
Day	Zinc %	no.	Control %	no.	Zinc %	no.	Control %	no.
0	100	69	100	67	100	68	100	71
1	79.7	55	86.6	58	98.5	67	98.6	70
2	59.4	41	73.1	49	89.7	61	94.4	67
3	36.2	25	53.7	36	80.8	55	81.7	58
4	21.7	15	28.4	19	72.1	49	67.6	48
5	18.8	13	23.8	16	57.3	39	56.3	40
6	14.5	10	22.4	15	44.1	30	43.7	31
7	13	9	21	14	36.7	25	38	27
	Turner 2000 (A-2) Fig.1A [19]				Turner 2000 (B-2) Fig.1B [19]			
Day	Zinc %	no.	Control %	no.	Zinc %	no.	Control %	no.
0	100	66	100	67	100	72	100	71
1	92.4	61	86.6	58	98.6	71	98.6	70
2	77.3	51	73.1	49	93.1	67	94.4	67
3	54.5	36	53.7	36	86.1	62	81.7	58
4	36.3	24	28.4	19	72.2	52	67.6	48
5	28.7	19	23.8	16	58.3	42	56.3	40
6	19.7	13	22.4	15	48.6	35	43.7	31
7	16.6	11	21	14	36.1	26	38	27
	Turner 2000 (A-3) Fig.1A [19]				Turner 2000 (B-3) Fig.1B [19]			
Day	Zinc %	no.	Control %	no.	Zinc %	no.	Control %	no.
0	100	70	100	67	100	68	100	71
1	81.4	57	86.6	58	100	68	98.6	70
2	65.7	46	73.1	49	92.6	63	94.4	67
3	50	35	53.7	36	83.8	57	81.7	58
4	38.6	27	28.4	19	67.6	46	67.6	48
5	25.7	18	23.8	16	51.5	35	56.3	40
6	18.6	13	22.4	15	38.2	26	43.7	31
7	12.8	9	21	14	28	19	38	27
	Turner 2000 (3-arms combined) Fig.1A [19]				Turner 2000 (3-arms combined) Fig.1B [19]			
Day	Zinc %	no.	Control %	no.	Zinc %	no.	Control %	no.
0	100	205	100	67	100	208	100	71
1	84.4	173	86.6	58	99.0	206	98.6	70
2	67.3	138	73.1	49	91.8	191	94.4	67
3	46.8	96	53.7	36	83.7	174	81.7	58
4	32.2	66	28.4	19	70.7	147	67.6	48
5	24.4	50	23.8	16	55.8	116	56.3	40
6	17.6	36	22.4	15	43.8	91	43.7	31
7	14.1	29	21	14	33.7	70	38	27

	Weismann 1990 Fig 1. [38]				
Day	Zinc %	no.	Control %	no.	
0	100	69	100	61	
1	98.5	68	100	61	
2	97.1	67	98.3	60	
3	92.8	64	93.4	57	
4	81.2	56	83.6	51	
5	65.2	45	68.9	42	
6	60.9	42	47.5	29	
7	49.3	34	31	19	

6.5 Mean days symptomatic, data extraction

	Zinc mean	SD	Control mean	SD	Data source
Belongia 2001 [30]	7.5	2.93	7.72	2.52	D'Cruze 2009 [13]
Douglas 1987 [39]	12.1	9.8	7.7	9.8	Hemilä 2011 [12]
Eby 1984 [27]	3.92	2.61	7.54	3.18	Hemilä 2011 [12]
Godfrey 1992 [40]	4.86	2.7	6.13	2.7	Hemilä 2017 [11]
Hirt 2000 [34]	2.3	0.9	9	2.5	Hirt 2000 [34]
Mossad 1996 [36]	5.2	2.83	9.2	5.32	Hemilä 2017 [11]
Mossad 2003 [35]	4.3	0.75	6	0.88	D'Cruze 2009 [13]
Petrus 1998 [31]	5.3	52	5.3	48	Petrus 1998 [31]
Prasad 2000 [29]	4.5	1.6	8.1	1.8	Prasad 2000 [29]
Prasad 2008 [28]	4	1.04	7.12	1.26	Prasad 2008 [28]
Smith 1989 [37]	7.23	2.29	7.57	3.01	Hemilä 2017 [11]
Turner 2000 (B-1) [19]	6.89	3.35	7.55	3.95	Hemilä 2011 [12]
Turner 2000 (B-2) [19]	7.9	4.25	7.55	3.96	Hemilä 2011 [12]
Turner 2000 (B-3) [19]	7.41	3.88	7.55	3.94	Hemilä 2011 [12]
Weismann 1990 [38]	7.16	2.62	6.72	2.29	Hemilä 2011 [12]

This was a secondary outcome

6.6 Incidence per-person month, data extraction & calculations

	Zinc events	Zinc total	Control events	Control total	Time months	Zinc Ee/Te	Control Ee/Te	IRR IRD	ln(IRR) ln(IRC)	SE ln(IRR) SE (IRD)
All RTIs										
Prasad 2007 [7]	7	24	19	25	12	0.024	0.063	0.384 -0.039	-0.958	0.442 0.017
Veverka 2009 [8]	5	20	4	17	7	0.036	0.034	1.063 0.002	0.061	0.671 0.023
Wei 2009 [22]	105	386	141	387	1	0.272	0.364	0.747 -0.092	-0.292	0.129 0.041
Zhang 2009 [23]	139	978	210	967	1	0.142	0.217	0.654 -0.075	-0.424	0.109 0.019
Mild RTIs										
Prasad 2007 [7]	7	24	16	25	12	0.024	0.053	0.456 -0.029	-0.786	0.453 0.016

Wei 2009 [22]	104	386	133	387	1	0.269	0.344	0.784 -0.074	-0.243	0.131 0.040
Zhang 2009 [23]	137	978	193	967	1	0.140	0.200	0.702 -0.060	-0.354	0.112 0.019
Moderate RTIs										
Prasad 2007 [7]	0.5	24	3	25	12	0.002	0.010	0.174 -0.008	-1.751	1.528 0.006
Wei 2009 [22]	1	386	8	387	1	0.003	0.021	0.125 -0.018	-2.077	1.061 0.008
Zhang 2009	2	978	17	967	1	0.002	0.018	0.116 -0.016	-2.151	0.748 0.005
Non-serious AEs										
Veverka 2009 [8]	0.5	20	2	20	7	0.004	0.014	0.250 -0.011	-1.386	1.581 0.011
Wei 2009 [22]	26	386	21	387	1	0.067	0.054	1.241 0.013	0.216	0.293 0.018
Zhang 2009 [23]	75	978	30	967	1	0.077	0.031	2.472 0.046	0.905	0.216 0.011

IRR: incidence rate ratio; **IRD:** incidence rate difference

$$SE(\log(IRR)) = \sqrt{\left(\frac{1}{a}\right) + \left(\frac{1}{b}\right)} \qquad SE(Rate\ Difference) = \sqrt{\frac{a}{(t_1)^2} + \frac{c}{(t_0)^2}}$$

6.1 Contour enhanced funnel plots: R code

6.1.1 Mean duration of symptoms (days) from community acquired mild RTI

Study	tx.n	tx.mean	tx.SD	con.n	con.mean	con.SD
Belongia 2001	81	7.5	2.93	79	7.72	2.52
Douglas 1987	33	12.1	9.8	30	7.7	9.8
Eby 1984	37	3.92	2.61	28	7.54	3.18
Godfrey 1992	35	4.86	2.7	28	6.13	2.7
Hirt 2000	108	2.3	0.9	105	9	2.5
Mossad 1996	49	5.2	2.83	50	9.2	5.32
Mossad 2003	40	4.3	0.75	38	6	0.88
Petrus 1998	52	5.3	2.9	49	7.1	4.2
Prasad 2000	25	4.5	1.6	23	8.1	1.8
Prasad 2008	25	4	1	25	7.1	1.3

Smith 1989	53	7.23	2.29	57	7.57	3.01
Weismann 1990	69	7.16	2.62	61	6.72	2.29

R code

```
> library(readxl)
> zinc_duration2 <- read_excel("Downloads/zinc_duration2.xlsx")
> View(zinc_duration2)
> zinc_duration=data.frame(zinc_duration2)
> zinc_meta=metacont(zinc_duration[,2], zinc_duration[,3],  zinc_duration[,4], zinc_duration[,5],
zinc_duration[,6], zinc_duration[,7], sm=SMD )
```

6.1.2 Adverse events for treatment of RTI: Zinc vs. placebo

study	events.int	n.int	events.con	n.con
Belongia 2001	41	81	40	78
Eby 1984	27	48	6	33
Eby 2006	11	25	8	22
Farr 1987 (B)	2	23	0	20
Godfrey 1992	20	35	15	38
Hemilä 2020	29	46	13	42
Hirt 2000	45	108	39	105
Mossad 1996	44	49	30	50
Mossad 2003	12	40	5	38
Turner 2001	21	41	21	50
Weismann 1990	21	61	15	69

R code

```
> Library(meta)
> library(readxl)
> Zinc_AE2 <- read_excel("Downloads/Zinc_AE2.xlsx")
> View(Zinc_AE2)
> zinc_AE=data.frame(Zinc_AE2)
> Zinc_meta_AE=metabin(zinc_AE[,2],zinc_AE[,3], zinc_AE[,4], zinc_AE[,5])
> cc <- funnel(Zinc_meta_AE, xlim=c(0.01,30),
          level = 0.95, contour = c(0.9, 0.95, 0.99))$col.contour
```

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